

Krylov-Subspace Model-Order Reduction for Fast Frequency Sweeps in the BP-FEM Solver

BP-FEM Optimization Track

Draft for review

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1 Scope

This document specifies the mathematical foundation of the Krylov-subspace Model-Order Reduction (MOR) module to be added to the BP-FEM solver as a *fast frequency sweep* option,

alongside the existing point-by-point direct solver and the BiCGStab fallback. The goal is to reduce a sweep of N_f frequency points over a band of interest from $\mathcal{O}(N_f)$ factorizations of the full system to a single (or few) factorization(s) plus $\mathcal{O}(N_f)$ evaluations on a reduced model whose dimension is two orders of magnitude smaller.

The document is intentionally implementation-aware: every symbol below has a matching counterpart in the existing C++ source tree (`FEMAssembler`, `PortModeSolver`, `MklPardisoSolver`, `ResultExtractor`, `SparseMatrix`). Where convenient, source-level names are noted in gray comments such as `// FEMAssembler::assemble`.

2 Continuous problem and FE discretization

2.1 Strong form

We solve the time-harmonic curl-curl problem in a bounded Lipschitz domain $\Omega \subset \mathbb{R}^3$ partitioned into wave-port faces $\{\Gamma_p\}_{p=1}^{N_p}$ and a perfect electric conductor (PEC) wall Γ_{PEC} with $\partial\Omega = \Gamma_{\text{PEC}} \cup \bigcup_p \Gamma_p$. The time convention is $e^{+i\omega t}$. For an angular frequency $\omega = 2\pi f$ the electric field $\mathbf{E}(\mathbf{r}, \omega) \in H(\text{curl}; \Omega)$ satisfies

$$\nabla \times (\mu_r^{-1} \nabla \times \mathbf{E}) - k_0^2 \varepsilon_c(\omega) \mathbf{E} = \mathbf{0} \quad \text{in } \Omega, \quad (1)$$

$$\hat{\mathbf{n}} \times \mathbf{E} = \mathbf{0} \quad \text{on } \Gamma_{\text{PEC}}, \quad (2)$$

$$\hat{\mathbf{n}} \times (\mu_r^{-1} \nabla \times \mathbf{E}) = \mathcal{B}_p(\mathbf{E}; \omega) + \mathbf{g}_p(\omega) \quad \text{on } \Gamma_p, \quad (3)$$

where $k_0 = \omega/c_0$, c_0 is the vacuum speed of light, and $\varepsilon_c(\omega) = \varepsilon_r - i\sigma/(\omega\varepsilon_0)$ encodes optional volumetric losses (`FEMAssembler::assemble`).

2.2 Wave-port boundary form

Each port Γ_p admits a numerically computed dominant transverse mode $\mathbf{e}_p^{(t)} : \Gamma_p \rightarrow \mathbb{R}^3$ with cutoff wavenumber $k_{c,p}$ and propagation constant $\beta_p(\omega) = \sqrt{k_0^2 - k_{c,p}^2}$, absorbing the propagating mode and exciting the projector $\mathbf{g}_p = -2i\beta_p a_p^{\text{inc}} (\hat{\mathbf{n}} \times \mathbf{e}_p^{(t)})$ where a_p^{inc} is the complex amplitude of the incident wave in $\sqrt{W}$ units. The L^2 -shape of $\mathbf{e}_p^{(t)}$ is normalized to $\int_{\Gamma_p} |\mathbf{e}_p^{(t)}|^2 dS = 1$ and the power normalization $s_p(\omega)$ is supplied by `powerNormalizationFactor`:

$$s_p(\omega) = \frac{1}{\sqrt{P_p^{\text{unit}}(\omega)}}, \quad P_p^{\text{unit}}(\omega) = \frac{\beta_p(\omega)}{2\omega\mu_0} \int_{\Gamma_p} |\mathbf{e}_p^{(t)}|^2 dS. \quad (4)$$

With this normalization the incident amplitude carried by the right-hand side becomes $\sqrt{P_p^W} s_p(\omega)$ where P_p^W is the requested incident power (`PortDefinition::magnitudeW`).

2.3 Galerkin discretization

Let $V_h \subset H(\text{curl}; \Omega)$ be the finite-element subspace spanned by H(curl)-conforming Nedelec/Whitney edge elements (0th or 1st-order hierarchical; `EdgeTopology`). Denote the global degree-of-freedom

vector $\mathbf{x}(\omega) \in \mathbb{C}^n$. The discrete bilinear forms and load vector at frequency ω read

$$\mathbf{K}_{ij} = \int_{\Omega} \mu_r^{-1} (\nabla \times \mathbf{w}_i) \cdot (\nabla \times \mathbf{w}_j) dV, \quad [\text{stiffness}] \quad (5)$$

$$\mathbf{M}_{ij} = \int_{\Omega} \varepsilon_c(\omega) \mathbf{w}_i \cdot \mathbf{w}_j dV, \quad [\text{mass; weakly } \omega\text{-dependent if } \sigma \neq 0] \quad (6)$$

$$\mathbf{m}_p = \int_{\Gamma_p} (\hat{\mathbf{n}} \times \mathbf{e}_p^{(t)}) \cdot \mathbf{w}_i dS, \quad [\text{port projector / coupling vector}] \quad (7)$$

$$\mathbf{f}(\omega) = \sum_{p=1}^{N_p} 2i\beta_p(\omega) \sqrt{P_p^W} s_p(\omega) \delta_p^{\text{exc}} \mathbf{m}_p, \quad [\text{wave-port RHS}] \quad (8)$$

which assemble into the discrete system

$$\boxed{\mathbf{A}(\omega) \mathbf{x}(\omega) = \mathbf{f}(\omega), \quad \mathbf{A}(\omega) = \mathbf{K} - k_0^2 \mathbf{M} + i \sum_{p=1}^{N_p} \beta_p(\omega) \mathbf{m}_p \mathbf{m}_p^\top.} \quad (9)$$

Equation (9) is the *exact* system the existing solver assembles per frequency (`FEMAssembler::assemble` for $\mathbf{K}, \mathbf{M}, \mathbf{m}_p$; `applyWavePorts` for the rank-one port term). The S-parameter postprocessor recovers

$$b_p(\omega) = \frac{\mathbf{m}_p^\top \mathbf{x}(\omega)}{s_p(\omega)}, \quad S_{pq}(\omega) = \frac{b_p(\omega) - a_p^{\text{inc}} \delta_{pq}}{a_q^{\text{inc}}} \quad (\text{for unit excitation } a_q^{\text{inc}} = 1). \quad (10)$$

Remark 1 (Why this form is MOR-friendly). *The system $\mathbf{A}(\omega)$ in (9) is a low-order rational matrix function of ω :*

- $\mathbf{K} - k_0^2 \mathbf{M}$ is a polynomial of degree 2 in k_0 , hence in ω , modulo a possibly weak ω -dependence of ε_c that is exactly handled by the model in §3.3.
- Each port term $i\beta_p(\omega) \mathbf{m}_p \mathbf{m}_p^\top$ is a rank-one update with frequency dependence concentrated in the scalar $\beta_p(\omega) = \sqrt{k_0^2 - k_{c,p}^2}$.

This structure is what allows a Krylov subspace built once at one (or a few) expansion points to capture the entire transfer function $S(\omega)$.

3 Linear time-invariant transfer model

3.1 State-space embedding

To keep the algebra clean we treat k_0^2 (or, for second-order moment matching, ω) as the expansion parameter and write the parametric system in descriptor form

$$[\mathbf{K} - k_0^2 \mathbf{M} + i \sum_p \beta_p(\omega) \mathbf{m}_p \mathbf{m}_p^\top] \mathbf{x} = \mathbf{B} \mathbf{u}, \quad \mathbf{y} = \mathbf{C} \mathbf{x}, \quad (11)$$

with input matrix $\mathbf{B} = [2i\beta_1 s_1 \mathbf{m}_1 \cdots 2i\beta_{N_p} s_{N_p} \mathbf{m}_{N_p}]$, input vector $\mathbf{u} \in \mathbb{C}^{N_p}$ holding the per-port incident amplitudes, and output matrix $\mathbf{C} = [\mathbf{m}_1 \cdots \mathbf{m}_{N_p}]^\top / \text{diag}(s_p)$. The transfer function of interest is

$$\mathbf{H}(\omega) = \mathbf{C} \mathbf{A}(\omega)^{-1} \mathbf{B} \in \mathbb{C}^{N_p \times N_p}, \quad (12)$$

whose (p, q) entry is precisely b_p/a_q^{inc} from (10) up to a known additive identity for the reflection diagonal.

3.2 Two MOR formulations

We support two formulations because they offer different tradeoffs and the choice is not premature optimization:

Form A — single-point “shifted-and-inverted” Krylov. Pick an expansion point ω_0 (band-center). Define the shifted operator $\mathbf{A}_0 = \mathbf{A}(\omega_0)$. The Krylov subspace

$$\mathcal{K}_q(\mathbf{A}_0^{-1}\mathbf{M}, \mathbf{A}_0^{-1}\mathbf{B}) = \text{span}\{\mathbf{A}_0^{-1}\mathbf{B}, (\mathbf{A}_0^{-1}\mathbf{M})\mathbf{A}_0^{-1}\mathbf{B}, \dots, (\mathbf{A}_0^{-1}\mathbf{M})^{q-1}\mathbf{A}_0^{-1}\mathbf{B}\} \quad (13)$$

matches $2q$ moments of $\mathbf{H}$ around ω_0 (block Arnoldi version, PRIMA-style).

Form B — multi-point (rational) Krylov. Choose a small set of expansion points $\{\omega_0^{(1)}, \dots, \omega_0^{(L)}\}$ inside the sweep band. For each expansion point factorize $\mathbf{A}_0^{(\ell)}$ once and append q_ℓ columns of the corresponding shift-and-invert Krylov subspace. The union spans a rational Krylov subspace that matches moments at every expansion point and is far more accurate over wide bands than Form A.

Both forms project $\mathbf{A}(\omega)$ onto the subspace $\mathbf{V}$ to produce the reduced model

$$\tilde{\mathbf{A}}(\omega) = \mathbf{V}^* \mathbf{A}(\omega) \mathbf{V}, \quad \tilde{\mathbf{B}} = \mathbf{V}^* \mathbf{B}, \quad \tilde{\mathbf{C}} = \mathbf{C} \mathbf{V}, \quad (14)$$

yielding the order- q reduced transfer function $\tilde{\mathbf{H}}(\omega) = \tilde{\mathbf{C}} \tilde{\mathbf{A}}(\omega)^{-1} \tilde{\mathbf{B}}$.

Theorem 1 (Moment-matching, single-point form). *Let $\mathbf{V} \in \mathbb{C}^{n \times q}$ be an orthonormal basis of $\mathcal{K}_q(\mathbf{A}_0^{-1}\mathbf{M}, \mathbf{A}_0^{-1}\mathbf{B})$. The Galerkin-type reduced model with $\tilde{\mathbf{A}}(\omega) = \mathbf{V}^* \mathbf{A}(\omega) \mathbf{V}$ matches the first $2q$ moments of the original transfer function $\mathbf{H}(\omega)$ around ω_0 (Grimme’s theorem). In particular, $\tilde{H}_{pq}^{(k)}(\omega_0) = H_{pq}^{(k)}(\omega_0)$ for all $k = 0, 1, \dots, 2q - 1$ and p, q .*

Remark 2 (Symmetry preservation). $\mathbf{K}, \mathbf{M}$ are real symmetric and $\mathbf{m}_p \mathbf{m}_p^\top$ are SPSD in this assembly. With Galerkin projection ($\mathbf{V}^*$ on the left, $\mathbf{V}$ on the right) and $\mathbf{V}$ orthonormal, the reduced system retains the same complex-symmetric structure, so the reduced $\tilde{\mathbf{A}}$ is a tiny dense complex-symmetric matrix solvable by LDL^T in $\mathcal{O}(q^3)$.

3.3 Frequency-dependent material model

Let $\varepsilon_c(\omega)$ be a rational function of ω . The generic bilinear-form parametric expansion is

$$\mathbf{A}(\omega) = \mathbf{K} + \sum_{i=0}^{N_a} \alpha_i(\omega) \mathbf{A}_i, \quad \alpha_i(\omega) \in \{1, k_0^2, ik_0, i\beta_p(\omega), \dots\}. \quad (15)$$

Each $\mathbf{A}_i$ is assembled *once* during the offline phase and stored in CSR alongside the existing `SparsePattern` cache. The reduced operator becomes

$$\tilde{\mathbf{A}}(\omega) = \mathbf{V}^* \mathbf{K} \mathbf{V} + \sum_i \alpha_i(\omega) \mathbf{V}^* \mathbf{A}_i \mathbf{V}, \quad (16)$$

so the online cost per frequency is $\mathcal{O}(N_a q^2)$ assembly plus $\mathcal{O}(q^3)$ solve, independent of the full DOF count n .

Algorithm 1 Block-Arnoldi (single-point Krylov MOR)

Require: expansion point ω_0 , max moments q , residual tolerance τ

Operators: $\mathbf{K}$, $\mathbf{M}$, $\mathbf{B}$, $\mathbf{C}$ already assembled

- 1: Form $\mathbf{A}_0 \leftarrow \mathbf{A}(\omega_0)$ and factorize once via PARDISO $\triangleright$ MklPardisoSolver: re-use symbolic factorization across frequencies for the reduction phase
 - 2: $\mathbf{V}_1 \leftarrow \text{orth}(\mathbf{A}_0^{-1} \mathbf{B})$
 - 3: **for** $j = 1, \dots, q$ **do**
 - 4: $\mathbf{W} \leftarrow \mathbf{A}_0^{-1} \mathbf{M} \mathbf{V}_j$
 - 5: **for** $i = 1, \dots, j$ **do**
 - 6: $\mathbf{H}_{ij} \leftarrow \mathbf{V}_i^* \mathbf{W}$, $\mathbf{W} \leftarrow \mathbf{W} - \mathbf{V}_i \mathbf{H}_{ij}$ $\triangleright$ Modified Gram-Schmidt
 - 7: **end for**
 - 8: (Re-orthogonalize $\mathbf{W}$ against $\mathbf{V}_{1:j}$ if needed.)
 - 9: $\mathbf{V}_{j+1} \leftarrow \text{orth}(\mathbf{W})$; break if $\|\mathbf{W}\| < \tau \|\mathbf{V}_1\|$
 - 10: **end for**
 - 11: $\mathbf{V} \leftarrow [\mathbf{V}_1, \mathbf{V}_2, \dots]$
 - 12: Build reduced matrices: $\tilde{\mathbf{K}} = \mathbf{V}^* \mathbf{K} \mathbf{V}$, $\tilde{\mathbf{M}} = \mathbf{V}^* \mathbf{M} \mathbf{V}$, $\tilde{\mathbf{m}}_p = \mathbf{V}^* \mathbf{m}_p$
 - 13: **return** $\mathbf{V}$, $\tilde{\mathbf{K}}$, $\tilde{\mathbf{M}}$, $\{\tilde{\mathbf{m}}_p\}$
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4 Algorithms

4.1 Block-Arnoldi for single-point form

4.2 Multi-point (rational) Krylov

5 Error and stability indicators

We expose three indicators; the user-facing tolerance is on (R3) but (R1) and (R2) are essential for diagnostics.

(R1) Algebraic residual. For an evaluated frequency the relative residual of the reduced solution lifted back to the full space is

$$\rho_{\text{alg}}(\omega) = \frac{\|\mathbf{A}(\omega) \mathbf{V} \tilde{\mathbf{x}}(\omega) - \mathbf{f}(\omega)\|_2}{\|\mathbf{f}(\omega)\|_2}. \quad (17)$$

If ρ_{alg} explodes the basis is too small or the expansion point is poorly placed. This indicator requires only one SpMV with the full system, no factorization.

(R2) Heuristic moment residual. Fully embedded in the ROM:

$$\rho_{\text{mom}}(\omega) = \|\tilde{\mathbf{A}}(\omega) \tilde{\mathbf{x}}(\omega) - \tilde{\mathbf{B}} \mathbf{u}\|_2 / \|\tilde{\mathbf{B}} \mathbf{u}\|_2. \quad (18)$$

This is essentially a self-consistency check on the reduced linear solve; useful for catching condition-number blowup of $\tilde{\mathbf{A}}$.

(R3) S-parameter discrepancy at validation points. Pick a small validation set $\Omega_v \subset [\omega_a, \omega_b]$ and run the existing direct solver. Define

$$\rho_S = \max_{\omega \in \Omega_v} \|\tilde{\mathbf{S}}(\omega) - \mathbf{S}(\omega)\|_F, \quad (19)$$

which is the user contract: the ROM output must agree with the exact solver to within $\rho_S \leq \varepsilon_S$ (default 0.01 dB on every $|S|$).

Algorithm 2 Adaptive rational Krylov MOR sweep

Require: sweep band $[\omega_a, \omega_b]$, target tolerance ε_S , max ROM order $q_{\max}$

- 1: Pick initial expansion points Σ_0 (e.g. $\{\omega_a, \omega_m, \omega_b\}$)
 - 2: $\mathbf{V} \leftarrow []$
 - 3: **for** each $\omega_0^{(\ell)} \in \Sigma_0$ **do**
 - 4: Factorize $\mathbf{A}_0^{(\ell)} = \mathbf{A}(\omega_0^{(\ell)})$
 - 5: Run Algorithm 1 for q_ℓ steps; append columns to $\mathbf{V}$
 - 6: **end for**
 - 7: Orthonormalize $\mathbf{V}$ globally; build reduced operators
 - 8: **repeat**
 - 9: Evaluate $\widetilde{\mathbf{H}}(\omega)$ on a probe set $\Omega_p \subset [\omega_a, \omega_b]$
 - 10: Compute residual indicator $\rho(\omega)$ from §5
 - 11: $\omega^* \leftarrow \arg \max_{\omega \in \Omega_p} \rho(\omega)$
 - 12: **if** $\rho(\omega^*) > \varepsilon_S$ and $\dim \mathbf{V} < q_{\max}$ **then**
 - 13: Factorize $\mathbf{A}(\omega^*)$, append Krylov block, re-orthonormalize
 - 14: **end if**
 - 15: **until** $\rho(\omega^*) \leq \varepsilon_S$ or budget exhausted
 - 16: **return** $\mathbf{V}$ and reduced operators
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6 Output reconstruction

6.1 S-parameter evaluation

For each requested ω :

1. Solve the dense reduced system $\widetilde{\mathbf{A}}(\omega) \widetilde{\mathbf{x}}(\omega) = \widetilde{\mathbf{B}} \mathbf{u}$ with the (cached) LDL^T of $\widetilde{\mathbf{A}}(\omega)$.
2. Form $\mathbf{y}(\omega) = \widetilde{\mathbf{C}} \widetilde{\mathbf{x}}(\omega)$.
3. Apply the existing power normalization $b_p = y_p$ (already divided by s_p inside $\widetilde{\mathbf{C}}$).
4. Compute S_{pq} exactly as in (10).

6.2 Field reconstruction

When a full-space field is requested at ω :

$$\mathbf{x}(\omega) \approx \mathbf{V} \widetilde{\mathbf{x}}(\omega). \quad (20)$$

The reduced model thus also provides VTU output at any sweep point at post-projection cost $\mathcal{O}(nq)$. Memory permitting, only requested frequencies need lifting; the default flag stays `-no-write-all-fields` to keep disk usage bounded.

7 Cost model

Let n = full DOFs, q = ROM dimension, N_f = sweep points, N_v = validation points, L = expansion points, $T_{\text{LU}}(n)$ = PARDISO factorization cost, $T_{\text{solve}}(n)$ = single triangular solve cost, N_p = port count.

For the BP-filter benchmark used throughout this project ($n \approx 2.9 \times 10^5$ order-1 DOFs, $N_f = 401$, $L = 2-3$, $q = 30-60$, $N_v \sim 5$): the speedup factor is $N_f/(L + N_v) \sim 401/8 \approx 50$, i.e. a $\sim 50\times$ wall-clock reduction is realistic before any parallelization.

Table 1: Asymptotic cost comparison for a sweep of N_f points.

Phase	Direct sweep	Krylov MOR sweep
Assembly	$N_f \cdot T_{\text{asm}}(n)$	$L \cdot T_{\text{asm}}(n) + \text{cached subforms}$
Factorization	$N_f \cdot T_{\text{LU}}(n)$	$L \cdot T_{\text{LU}}(n)$
Solve / Krylov build	$N_f \cdot N_p \cdot T_{\text{solve}}(n)$	$L \cdot q \cdot N_p \cdot T_{\text{solve}}(n)$
Per-frequency online	0	$\mathcal{O}(N_a q^2 + q^3)$
Validation	0	$N_v \cdot T_{\text{LU}}(n)$

8 Numerical safeguards

- **Re-orthogonalization.** Modified Gram-Schmidt with one or two reorthogonalization passes; switch to Householder if loss of orthogonality exceeds 10^{-10} .
- **Deflation.** Drop columns whose norm after orthogonalization drops below 10^{-12} ; report deflation events.
- **Conditioning.** Monitor $\|\tilde{\mathbf{A}}^{-1}\|/\|\tilde{\mathbf{A}}\|$ via LDL^T pivot magnitudes. Trigger an additional expansion point if conditioning exceeds 10^{12} .
- **Cutoff handling.** Below the lowest-order port cutoff $k_0 < k_{c,p}$, β_p becomes imaginary and $s_p \rightarrow 0$. The reduced model must extend continuously by analytic continuation; we keep β_p in the symbolic form $\sqrt{k_0^2 - k_{c,p}^2}$ and branch-cut on the principal sqrt. The behavior is identical to the existing direct solver (`powerNormalizationFactor` returns 0).
- **Symmetry recovery.** If round-off perturbs symmetry, symmetrize $\tilde{\mathbf{K}} \leftarrow \frac{1}{2}(\tilde{\mathbf{K}} + \tilde{\mathbf{K}}^\top)$ explicitly after construction.

9 Mapping to existing code

Mathematical object	Source location
$\mathbf{K} - k_0^2 \mathbf{M}$	<code>src/fem/FEMAssembler.cpp::assemble</code> (volume part)
$i\beta_p \mathbf{m}_p \mathbf{m}_p^\top$	<code>src/fem/FEMAssembler.cpp::applyWavePorts</code>
$\mathbf{m}_p$ (output)	<code>src/fem/PortModeSolver.cpp::computeMode::couplingWeights</code>
$s_p(\omega)$	<code>src/fem/PortModeSolver.cpp::powerNormalizationFactor</code>
b_p, S_{pq}	<code>src/post/ResultExtractor.cpp::extract</code> / <code>portProjection</code>
PARDISO LU/LDL ^T of $\mathbf{A}(\omega_0)$	<code>src/linalg/MklPardisoSolver.cpp</code>
CSR pattern reuse	<code>src/linalg/SparseMatrix.cpp::createPattern,</code> <code>FEMAssembler::sparsityPattern_</code>

The MOR module will introduce a new namespace `fem::mor` living under `src/mor/` with no upward dependency on `app`; its only inputs are the existing `FEMAssembler`, `PortModeSolver`, and a `LinearSolver` abstraction.

10 Validation strategy

1. **Analytic check.** Empty rectangular waveguide with a flat lossless body: $S_{21}(\omega) = e^{-i\beta L}$ exactly. ROM with $L = 2$, $q = 20$ must match within 10^{-6} in $|S|$ over a 2:1 band.

2. **Loaded BP-filter (project benchmark).** The Krylov ROM with $L = 3$, $q = 40$ over 40–43 GHz vs. the direct 401-point sweep: $\rho_S \leq 0.05$ dB everywhere.
3. **Energy conservation.** For lossless cases verify $|S_{11}|^2 + |S_{21}|^2 \leq 1 + 10^{-3}$ at all sweep points after ROM evaluation (already a regression target for direct sweep).
4. **Convergence in q .** Plot $\max_{\omega} \|\tilde{\mathbf{S}} - \mathbf{S}\|_F$ vs. q at fixed L . Expect exponential decay until q saturates.

11 Limitations and out-of-scope items

- **Strong dispersion.** The current model handles linear σ losses exactly and Debye/Drude models via affine expansion. Highly non-rational $\varepsilon(\omega)$ (e.g. measured tabulated data with sharp resonances) requires either piecewise rational fitting or a different reduction method (e.g. AAA + balanced truncation), which is out of scope.
- **Strong nonlinearity.** Out of scope; non-iterative MOR is not meaningful here.
- **Eigenmode tracking near band edges.** ROM reproduces the transfer function but does not robustly identify resonances inside stop-bands; a separate Krylov-Schur eigenvalue path is delegated to a future module.
- **Adaptive p-refinement coupling.** ROM is built on a fixed mesh and basis order. h/p-adaptivity in the same sweep changes operators between iterations; supporting this requires re-projecting $\mathbf{V}$ onto the new V_h , which is treated in a future revision.

12 References (informal)

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